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Programming Challenges.

Programming Challenges Question - 1

Write a program that accepts 10 student records (roll number and score) and prints them in decreasing order of scores. In case there are multiple records pertaining to the same student, the program should choose a single record containing the highest score. The program should be capable of accepting a multi-line input. Each subsequent line of input will contain a student record, that is, a roll number and a score (separated by a hyphen). The output should consist of the combination of roll number and corresponding score in decreasing order of score. INPUT to program

```

1001-40
1002-50
1003-60
1002-80
1005-35
1005-55
1007-68
1009-99
1009-10
1004-89

```

OUTPUT from program

```

1009-99
1004-89
1002-80
1007-68
1003-60
1005-55
1001-40

```

Solution of Challenge - 1

Programming Challenges Question - 2

Sam wants to select a username in order to register on a website. The rules for selecting a username are:

1. The minimum length of the username must be 5 characters and the maximum may be 10.
2. It should contain at least one letter from A-Z
3. It should contain at least one digit from 0-9
4. It should contain at least one character from amongst @#*=
5. It should not contain any spaces

Write a program which accepts 4 usernames (one username per line) as input and checks whether each of them satisfy the above mentioned conditions.

If a username satisfies the conditions, the program should print PASS (in uppercase)

If a username fails the conditions, the program should print FAIL

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(in uppercase)

Suppose the following usernames are supplied to the program:

```
1234@a
ABC3a#@
1Ac@
ABC 3a#@
```

Then the output should be:

```
PASS
PASS
FAIL
FAIL
```

[Solution of Challenge - 2](#)

Programming Challenges Question - 3

Kermit, a frog hops in a particular way such that:

1. He hops 20cm in the first hop, 10cm in the second hop and 5cm in the third hop.
2. After three hops Kermit rests for a while and then again follows the same hopping pattern.

Calculate the total distance traveled by Kermit (in centimeters) for the provided number of hops. Exactly 4 numbers of hops will be provided to the program (one number per line) as per the below example.

Suppose the following number of hops is provided to the program:

```
4
6
3
5
```

Then the total distance covered should be displayed as follows:

```
55
70
35
65
```

[Solution of Challenge - 3](#)

Programming Challenges Question - 4

Write a program which will accept a single pair of strings separated by a comma; the program should calculate the sum of ascii values of the characters of each string. The program should then subtract the sum of the ascii values of the second string from the sum of the ascii values of the first string.

Suppose the following input is given to the program:

```
123ABC,456DEF
```

Then the sum of the ascii values of the characters in '123ABC' is 348 and in '456DEF' it is 366. The Difference between these numbers is $348 - 366 = -18$

The corresponding output to be printed by the program is:

```
-18
```

[Solution of Challenge - 4](#)

Programming Challenges Question - 5

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Write a program which will take the year (yyyy) and the numeric sequence of the month (0-11) as its input. The program will return the day on which the 28th of that particular month and year falls. The input can consist of two year-month combinations, one combination per line.

The numeric sequence of months is as follows:

0 - Jan
1 - Feb
2 - March
and so on.....

The format for supplying the input is:

1999-5

Where 1999 is the year and 5 is the numeric sequence of the month (corresponding to June). The program should display the day on which June 28, 1999 fell, and in this case the output will be MONDAY.

The output should be displayed in uppercase letters.

Suppose the following INPUT sequence is given to the program:

1999-5

1998-6

Then the output should be:

MONDAY

TUESDAY

[Solution of Challenge - 5](#)

Programming Challenges Question - 6

Write a program which will accept three sentences (one sentence per line) and print the words having Initial Caps within the sentences. Below is an example.

Here is an example. If the below three sentences are given to the program as

input,

This is a Program

Coding test of Initial Caps

the program Will Test You

Then, the output would look like:

This

Program

Coding

Initial

Caps

Will

Test

You

[Solution of Challenge - 6](#)

Programming Challenges Question - 7

Write a program which will print the below structures according to the input provided to the program. The program should accept 3 inputs in the form of numbers between 1

and 9, both inclusive (one number per line) and then generate the corresponding structures based on the input.

Suppose the following sequence of numbers is supplied to the program:

```
3
2
4
```

Then the output should be:

```
1
2 2
3 3 3
```

```
1
2 2
```

```
1
2 2
3 3 3
4 4 4 4
```

[Solution of Challenge - 7](#)

Programming Challenges Question - 8

Write a Java program to reverse a String in given particular way without creating any new String object.

Input- "Kushwaha Umesh is name My"

Output- "My name is Umesh Kushwaha"

[Solution of Challenge - 8](#)

Programming Challenges Question - 9

Write a method that given a list of non negative integers, arranges them such that they form the largest possible number. For example, given [50, 2, 1, 9], the largest formed number is 95021.

Input- [62, 5, 1, 3]

Output- 62531

[Solution of Challenge - 9](#)

Programming Challenges Question - 10

Write a program that outputs all possibilities to put + or - or nothing between the numbers 1, 2, ..., 9 (in this order) such that the result is always 100.

For example: $1 + 2 + 34 - 5 + 67 - 8 + 9 = 100$.

Input- TARGET_SUM = 100;

VALUES = { 1, 2, 3, 4, 5, 6, 7, 8, 9 };

Output-

1+2+3-4+5+6+78+9

```
1+2+3+4+5+6+7+8+9
1+2+3+4+5+6+7+8+9
1+2+3+4+5+6+7+8+9
1+2+3+4+5+6+7+8+9
1+2+3+4+5+6+7+8+9
1+2+3+4+5+6+7+8+9
1+2+3+4+5+6+7+8+9
1+2+3+4+5+6+7+8+9
1+2+3+4+5+6+7+8+9
1+2+3+4+5+6+7+8+9
```

Solution of Challenge - 10

Programming Challenges Question - 11

Write a program that accepts a comma separated string of numbers depicting the daily stock price of AXY Company. The program should calculate the biggest daily fluctuation (in terms of percentage) of the stock price on any given day.

For example, let us suppose the following is the input string:

INPUT

20,23,25,30,22,21,16

Then the corresponding biggest fluctuation is:

OUTPUT

26.6666666666667 (As the maximum fluctuation in price is between 22 and 30)

Solution of Challenge - 11

Programming Challenges Question - 12

Write a program that accepts 4 words as input (one per line) and identifies whether they are palindromes. A palindrome is a word that reads the same when read back to front. The program should print the alphabetically sorted list of palindromes (one per line). The program output should be in lowercase.

INPUT

waas

gaag

haaha

ABBA

OUTPUT

abba

gaag

Solution of Challenge - 12

Programming Challenges Question - 13

Write a program which will take the volume of the cube as its input. The program should calculate the area of a square whose side is half the length of the side of the cube. The program should print the area of the square.

Definitions:

Volume of cube = a^3 , where a is the side of a cube

Area of Square = b^2 , where b is the side of the square

For example, let us suppose the following is the volume inputted to the program:

INPUT

8

27

Then the corresponding output will be as follows:

OUTPUT

1
2.25

[Solution of Challenge - 13](#)

Programming Challenges Question - 14

Write a program which will accept a number as its input. You are required to calculate the sum of the first and last digits of the square of the number given as input. In case the square of the number is a single digit number then the square should be printed as it is.

For example, let us suppose the following input is given to the program:

INPUT 3 5 6

Then the corresponding output should be:

OUTPUT 9 7 9

[Solution of Challenge - 14](#)

Programming Challenges Question - 15

Write a program that accepts a number of ASCII values between 40 and 125 (both inclusive) and concatenates the corresponding characters to form a string. The program accepts a multi-line input. The first line contains a number representing how many ASCII values are in the input. The subsequent lines contain the ASCII values themselves (one ASCII value per line).

The following is an example of a valid input:

INPUT 5 (this indicates the total number of ASCII values in the input set) 65 67 69 70 71

The output is printed as a string. The ASCII value of A is 65, C is 67, E is 69, F is 70, G is 71 and therefore the output will be as follows:

OUTPUT
ACEFG

[Solution of Challenge - 15](#)

Programming Challenges Question - 16

During a car rally the time keeper wrote down the driver names along with their finishing positions in random order. Write a program to sort the names of the drivers in ascending order of their finishing positions.

For example, let us suppose the following input is given to the program:

INPUT Eddie-3 Michael-1 Nica-4 Montoya-6

Then the program should print the output as follows:

OUTPUT Michael-1 Eddie-3 Nica-4 Montoya-6

[Solution of Challenge - 16](#)

Programming Challenges Question - 17

Write a program which will accept a set of numbers. For each number, it will calculate the sum of the cubes for the first 'n' natural numbers counting up to that number. The first line of the input should consist of a number representing how many

numbers are in the input set. The subsequent lines contain the numbers themselves (one per line).

Here is an example of how the input and output would look:

INPUT3 (this indicates the total count of numbers in the input set) 1 2 4

OUTPUT1 9 100

[Solution of Challenge - 17](#)

Programming Challenges Question - 18

A bank requires its clients to change their internet banking passwords based on the following conditions:

Alphabets between A-Z or a-z (inclusive) are allowed

The number of alphabets should not be less than 4

Numbers between 0-9 (inclusive) are allowed

The count of numbers in the password should not be less than 2

Write a program to validate the internet banking password for the above conditions. If the password meets the conditions then "VALID" should be printed, else "INVALID" should be printed.

INPUTa2 abcdeF12 abcdeF12(*)

OUTPUTINVALID VALID VALID

[Solution of Challenge - 18](#)

Programming Challenges Question - 19

Write a program that evaluates the sum of all the divisors of a given number, other than the number itself. For example, if the number is 14, the divisors are 1,2,7 and 14. If we exclude 14, the sum of the remaining divisors will be $1+2+7 = 10$.

The program accepts a multi-line input. The first line contains a number representing how many numbers are being inputted.

Subsequent lines contain the numbers themselves. The inputted numbers should be between 1 and 100000 (inclusive). The output should be printed one per line corresponding to the sum of the divisors of the particular inputted number.

Below is some example input and output:

INPUT4 (this represents how many numbers exist in the input set)
13 15 10 6

OUTPUT1 9 8 6

Note: In case of input data being supplied to the question, it should be assumed to be a console input.

[Solution of Challenge - 19](#)

Programming Challenges Question - 20

Write a program which will convert the given RGB input values into the corresponding hexadecimal values preceded with a # sign.

A single input sequence will be provided, consisting of RGB values separated by "-". Each RGB combination will be separated by ",". The outputs should be comma separated and in uppercase. Also, it should be checked if a color value is greater than 255. In such a case, the output for the corresponding combination should be given as INVALID.

Note: Number 10, which has a hexadecimal value of A, must be represented as "0A" and not as "A". The same rule applies to other single digit hexadecimal numbers.

Example

Suppose the following input sequence is supplied to the program:

```
INPUT12-13-14,45-156-23,234-234-256
```

```
OUTPUT#0C0D0E,#2D9C17,INVALID
```

[Solution of Challenge - 20](#)

Programming Challenges Question - 21

Write a program to calculate the bill amount, in cents, for the units of power consumed. Following are the rates applicable:

First 0-100 units: 60 cents per unit

Next 200 units: 70 cents per unit

Beyond 300 units: 80 cents per unit

The program should accept three different usage unit readings.

Example

If the following inputs are supplied:

```
INPUT305 180 120
```

Then, the output should be:

```
OUTPUT20400 11600 7400
```

[Solution of Challenge - 21](#)

Programming Challenges Question - 22

Ross is an event organizer. He has received data regarding the participation of employees in two different events. Some employees have participated in only one event and others have participated in both events. Ross now needs to count the number of employees who have taken part in both events. The records received by Ross consist of employee ids, which are unique. Write a program that accepts the employee ids participating in each event (the first line relates to the first event and the second line relates to the second event). The program should print the number of common employee ids in both the events.

```
INPUT1001,1002,1003,1004,1005 1106,1008,1005,1003,1016,1017,1112
```

Now the common employee ids are 1003 and 1005, so the program should give the output as:

```
OUTPUT2
```

[Solution of Challenge - 22](#)

Programming Challenges Question - 23

Write a program to calculate the distance travelled by a car at different time intervals. The initial velocity of the car is 36 km/hr and the constant acceleration is 5 m/s².

The formula to calculate distance is:

Distance Travelled = $ut + (at^2)/2$ where, u = initial velocity of the car (36 km/hr) a = acceleration of the car (5 m/s²) t = time duration in seconds

The program should accept 2 time intervals as the input (one time

interval per line) and print the distance travelled in meters by the car (one output per line).

Definitions: 1 kilometer = 1000 meters 1 hour = 3600 seconds
Let us suppose following are the inputs supplied to the program
INPUT10 8
OUTPUT350 240

[Solution of Challenge - 23](#)

Programming Challenges Question - 24

Write a program that accepts 5 sentences as input (one sentence per line) and counts the number of vowels (a,e,i,o,u,A,E,I,O,U) occurring in each sentence. The program should print the vowel count corresponding to each of the five sentences as shown in the below example.

INPUTThis is a test The Fox and the Hound May the force be with
you Are you talking to me Go West
OUTPUT4 6 8 8 2

[Solution of Challenge - 24](#)

Programming Challenges Question - 25

Write a program which prints the numbers between 2 and 20000 (both inclusive) whose sixth root is an integer and which are a square of a number and also a cube of a number. An example of such a number is 64 since its sixth root is 2. 64 is also the square of 8 and cube of 4.

OUTPUTNum1 Num2 Num3

[Solution of Challenge - 25](#)

Programming Challenges Question - 26

Write a program which will validate four inputted ip addresses (one per line). A valid ip address must follow the form xxx.xxx.xxx.xxx where xxx is a number from 0-255. The program should print "VALID" if the ip address is valid, and "INVALID" if the ip address is invalid.

Example:

INPUT12.12.13.14. 256.245.234.221 1.2.3.4 23.24.25.26.

OUTPUTINVALID INVALID VALID INVALID

[Solution of Challenge - 26](#)

Programming Challenges Question - 27

Write a program which accepts 4 integer values (one per line) and converts them to their corresponding 16 bit binary values. The output should be printed one value per line.

INPUT8 10 17 23

OUTPUT00000000000001000 0000000000001010 0000000000010001
0000000000010111

[Solution of Challenge - 27](#)

Programming Challenges Question - 28

Write a program that accepts four dates (one per line) and checks to see if they are in the date format "mm-dd-yyyy" (month-day-year). If the date is found to be in the valid format, the program should print "VALID", if not, the program should print

"INVALID".Example:

INPUT01-01-1999 13-11-1999 10-30-1999

OUTPUTVALID INVALID VALID

[Solution of Challenge - 28](#)



Posted by [Umesh Kushwaha](#)

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You are the owner of a railway system between n cities, numbered by integers from 1 to n. Each train travels from the start station to the end station according to a very specific timetable (always on time), not stopping anywhere between. On each station a departure timetable is available. Unfortunately each timetable contains only direct connections. A passenger that

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